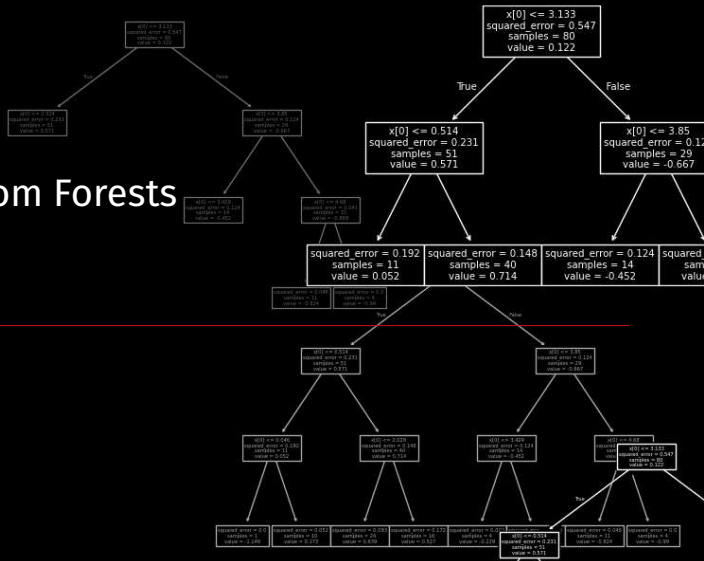


Decision trees and Random Forests

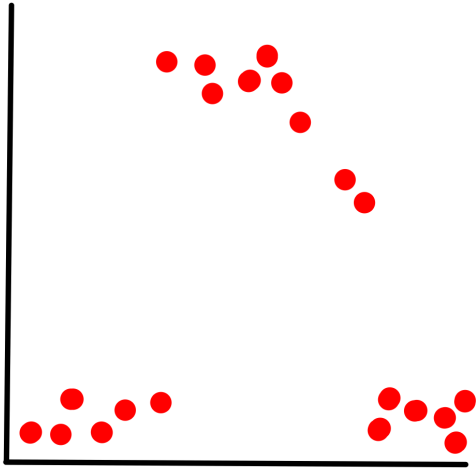
AI for ecologists

Paul Tresson

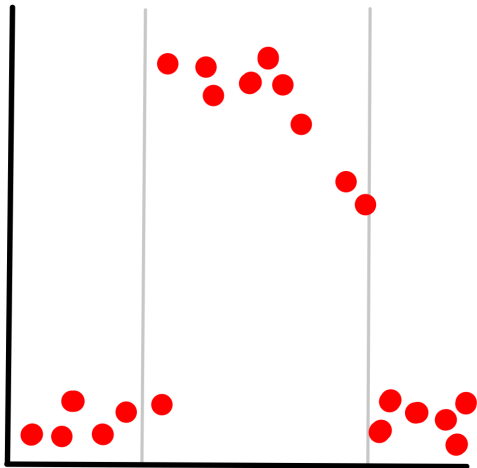


Introduction

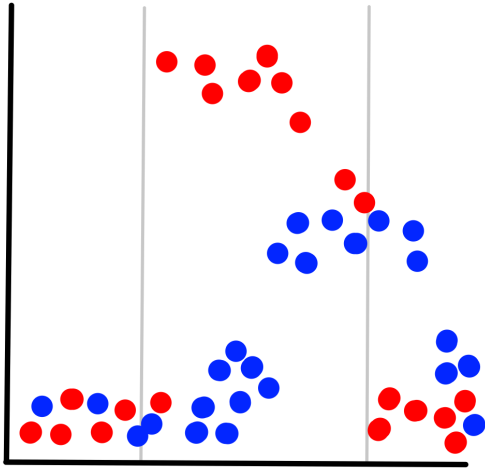
Motivation



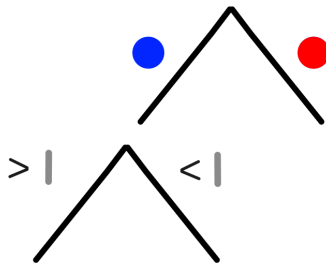
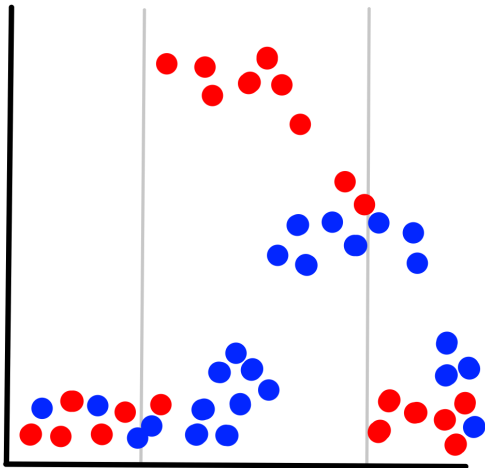
Motivation



Motivation



Motivation



Decision Trees

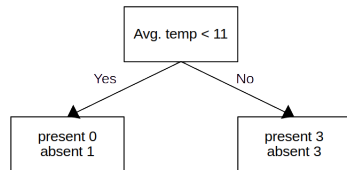
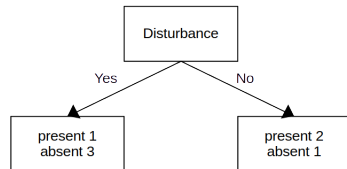
Simple example

Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
Yes	Forests	12	0
No	Shrubs	18	1
No	Shrubs	25	1
Yes	Shrubs	28	1
Yes	Forests	30	0
No	Forests	33	0

Adapted from StatQuest

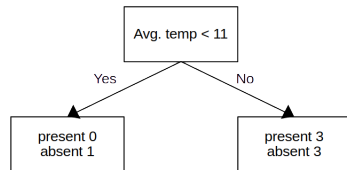
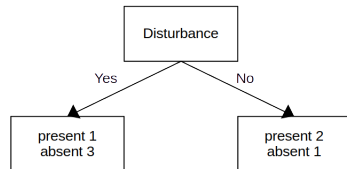
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No	Forests	33	0



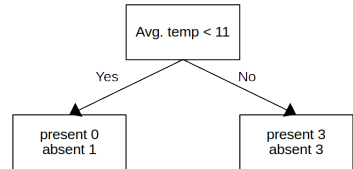
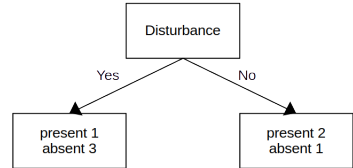
Gini impurity

$$\sum_{i=1}^J \left(p_i \sum_{k \neq i} p_k \right) = 1 - \sum_{i=1}^J p_i^2$$



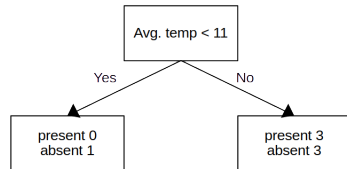
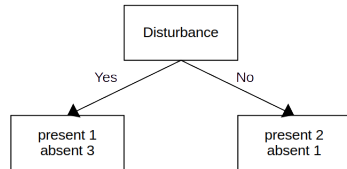
Gini impurity

$$1 - \left(\frac{1}{1+3}\right)^2 - \left(\frac{3}{1+3}\right)^2 = 0.375$$



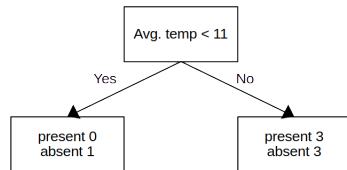
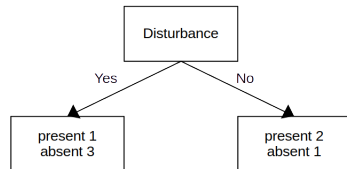
Gini impurity

$$\text{Leaf Gini} = \left(\frac{4}{4+3}\right)0.375$$



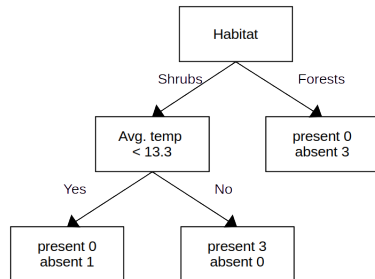
Gini impurity

$$1 - \left(\frac{0}{0+1}\right)^2 - \left(\frac{1}{0+1}\right)^2 = 0$$

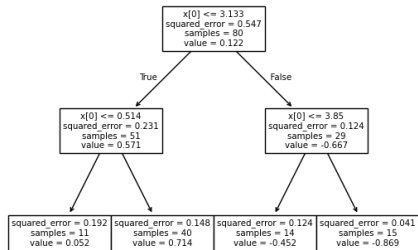
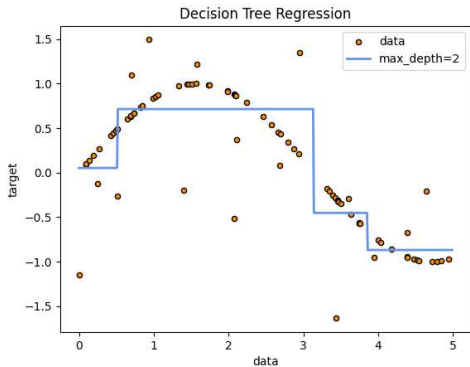


Building the tree

Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
Yes	Forests	12	0
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No	Shrubs	25	1
Yes	Shrubs	28	1
Yes	Forests	30	0
No	Forests	33	0

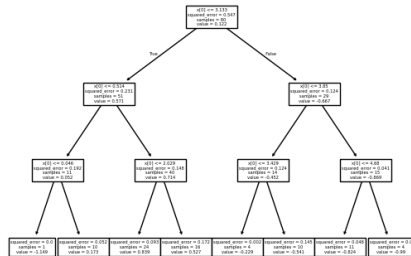
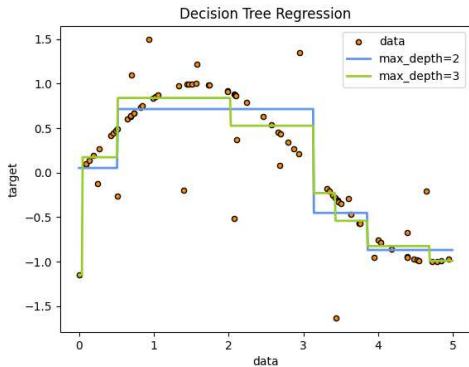


Regression trees



Adapted from sklearn documentation

Regression trees



Adapted from sklearn documentation



Regression trees

Non-linear data, multiple outputs !

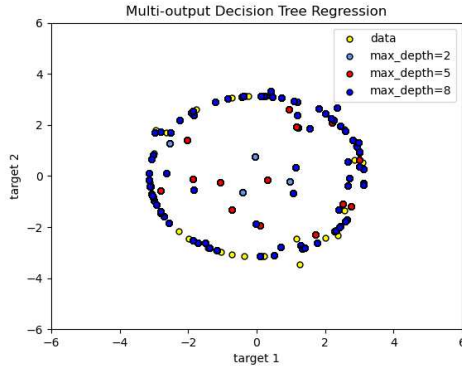


Figure from sklearn documentation

Random Forests

Main idea

Why not several trees ?

Boostraping

Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
Yes	Forests	12	0
No	Shrubs	18	1
No	Shrubs	25	1
Yes	Shrubs	28	1
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Boostraping

Disturbance	Habitat	Avg. temp.	Presence
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Yes	Forests	12	0
Yes	Forests	12	0
No	Shrubs	18	1
No	Shrubs	25	1
Yes	Shrubs	28	1
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Subset variables

Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
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Yes	Forests	12	0
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Yes	Shrubs	28	1
Yes	Shrubs	28	1

Subset variables

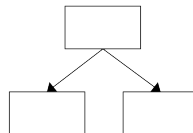
Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
Yes	Forests	12	0
Yes	Forests	12	0
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Yes	Forests	12	0
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No	Shrubs	25	1
Yes	Shrubs	28	1
Yes	Shrubs	28	1

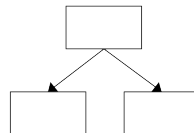
Building a tree

Disturbance	Habitat	Avg. temp.	Presence
Yes	Shrubs	10	0
Yes	Forests	12	0
Yes	Forests	12	0
No	Shrubs	18	1
No	Shrubs	25	1
Yes	Shrubs	28	1
Yes	Shrubs	28	1



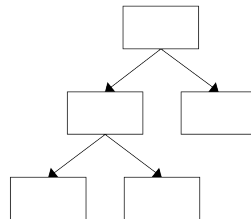
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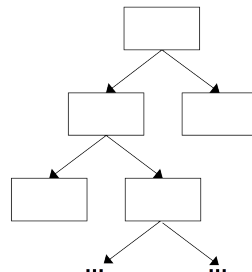
Building a tree

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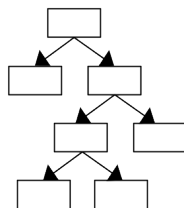
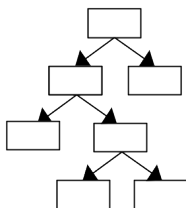
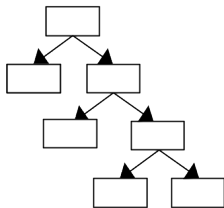
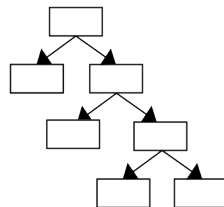
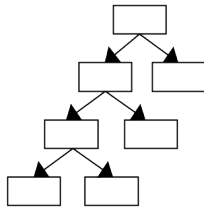
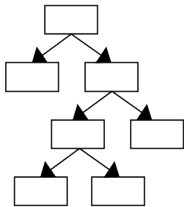


Building a tree

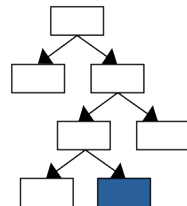
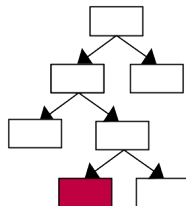
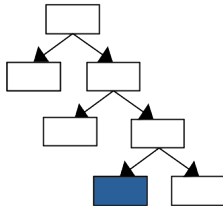
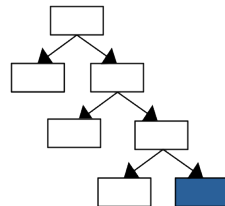
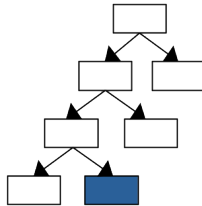
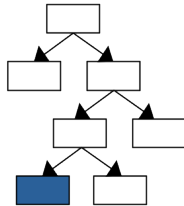
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Building a Forest



Using the Forest



RF advantages

- different inputs

RF advantages

- different inputs
- different outputs

RF advantages

- different inputs
- different outputs
- $\approx$ explainable

RF advantages

- different inputs
- different outputs
- $\approx$ explainable
- pretty easy to fit

RF advantages

- different inputs
 - different outputs
 - $\approx$ explainable
 - pretty easy to fit
- **seasoned and reliable**

RF drawbacks

- need to test hyper-parameters

RF drawbacks

- need to test hyper-parameters

How many trees ? how many subsets ? what depth ?

RF drawbacks

- need to test hyper-parameters

How many trees ? how many subsets ? what depth ?

- need for rich descriptors

Decendants and variants

- Adaboost
- Gradient Boosting
- XGBoost
- ...

Useful ressources

- `scikit-learn` docs !
- StatQuest

Thanks for you attention !